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Assignment -01

01) Discuss the main characteristics of the database approach and how it differs from traditional file systems.

→ The main characteristics of database approach are as follows

i) self-describing nature of a database system

- The database system contains not only database itself but also a complete definition or description of the database structure and constraints.
- The definition is stored in the DBMS catalog, which contains information such as the structure of each file, the type and storage format of each data item, and various constraints on the data.
- The information stored in the catalog is called meta-data, and it describes the structure of the primary database.
- A general purpose DBMS software package is not written for a specific database application.

ii) Insulation between programs and data, and data abstraction

- In DBMS access programs do not require such changes as the structure of data file is stored in the DBMS catalog separately from the access programs. This property is called program-data independence.
- User application programs can operate on the data by invoking these operations through their names and arguments, regardless of how the operations are implemented. This property is termed program-operation Independence.
- The characteristic that allows program-data independence and program-operation independence is called data abstraction.

iii) Support of Multiple Views of the Data

- A database typically has many users, each of whom may require a

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different perspective or view of the database.

- A view may be subset of the database or it may contain virtual data that is derived from the database files but is not explicitly stored.
- A multiuser DBMS whose users have a variety of distinct applications must provide facilities for defining multiple views.

iv) Sharing of Data and Multiuser Transaction processing

- A multiuser DBMS must allow multiple user to access the database at the same time.
- The DBMS must include concurrency control software to ensure that several user trying to update the same data do so in a controlled manner so that the result of the update is correct.
- For example, when several reservation agents try to assign a seat on an airline flight, the DBMS should ensure that each can be accessed by only one agent at a time for assignment to a passanger. These types of applications are generally called online transaction processing (OLTP) applications.
- The isolation property ensures that each transaction appears to execute in isolation from other transactions.
- The atomicity property ensures that either all the database operations in a transaction are executed or none are.

Difference b/w Database Approach and File system

Database Approach	File system
> Data redundancy is controlled.	> High Data redundancy.
> High Data consistency.	> Low consistency.
> supports multiple users.	> Limited sharing.
> strong security mechanisms.	> Weak security.
> Backup and recovery available.	> Difficulty to backup.

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> Centralized control.	> Decentralized control.
> Concurrency control present.	> Not supported.
> Uses query language (SQL)	> NO standard query language.

02) Discuss the advantages of DBMS over the traditional file system.

→ Advantages of using a DBMS are

i) Controlling Redundency
ii) Restricting Unauthorized access
iii) Providing Persistent Storage for program Objects and Data structures
iv) Permitting Inferencing and Actions using Rules
v) Providing Multiple user Interfaces
vi) Representing complex Relationship among Data.
vii) Enforcing Integrity constraints.
viii) Providing Backup and Recovery
ix) Additional Implications of the Database approach

i) Controlling Redundency
> storing the same data multiple times is called redundency.
> Redundency leads to serval problems like:
• Duplication of efforts • storage space is wasted • Inconsistency
> In some cases it is sometime necessary to use controlled redundancy to improve the performance of queries.

ii) Restricting unauthorized access
> A DBMS should provide a security & unauthorized access or subsystem, which the DBA uses to create accounts and to specify account restrictions.
> Parametric users may be allowed to access the database only through the predefined canned transactions developed for their use.

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iii) Providing persistent storage for Program objects
> Databases can be used to provide persistent ~~storage~~ storage for program objects and data structures.
> This is one of the main reasons for object-oriented database system
> object-oriented database system typically offer data structure compatibility with one or more OOPS languages

iv) Providing storage structures and search
iv) Permitting Inferencing and Actions using Rules
> some database system provides capabilities for defining deduction rules for inferencing new information from the stored database facts. such systems are called deductive database systems.
> More powerful functionality is provided by active database system.

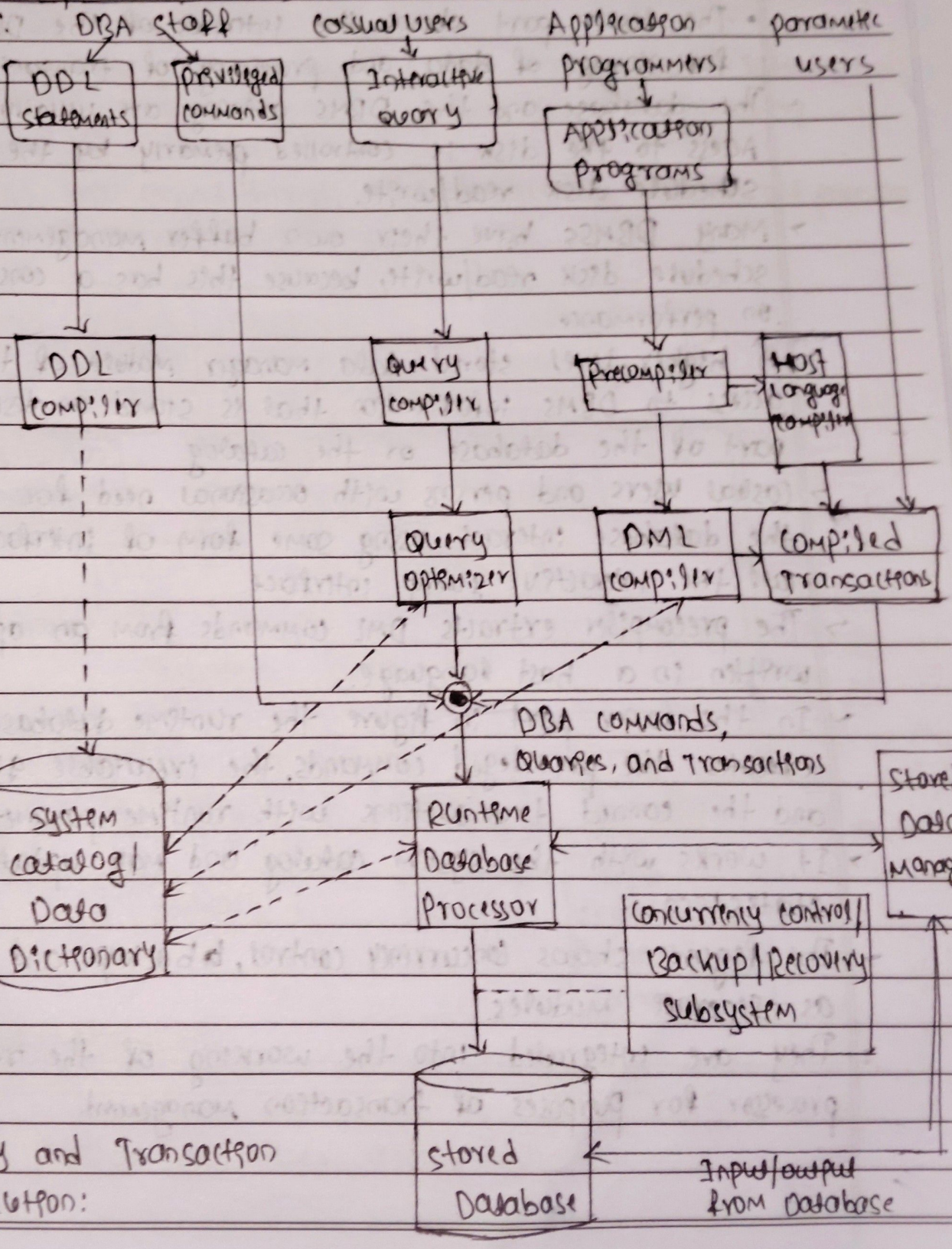
v) Providing Multiple user Interfaces
> Many types of users with varying levels of technical knowledge use a database, a DBMS should provide a variety of user interfa
> Ex: query languages for casual users.
> Both forms-style interfaces & Menu-driven interfaces are commonly known as graphical user interfaces (GUIs).

vi) Representing complex Relationship among data.
> A database may include numerous varieties of data that are interrelated in many ways.

vii) Enforcing Integrity constraints
> Most database applications have certain integrity constraints that must hold for the data.
> A DBMS should provide capabilities for defining and enforcing these constraints

03) Explain the component modules of DBMS and their interaction with the help of neat diagram

→ DBMS component modules



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> The figure is divided into two parts:
- The top part of the figure refers to the various users of the database environment and their interfaces.
- The lower part shows the internals of the DBMS responsible for storage of data and processing of transactions.

> The database and the DBMS catalog are usually stored on disk. Access to the disk is controlled primarily by the OS, which schedule disk read/write.

> Many DBMSs have their own buffer management module to schedule disk read/write, because this has a considerable effect on performance.

> A higher-level stored data manager module of the DBMS controls access to DBMS information that is stored on desk, whether it is part of the database or the catalog.

> Casual users and persons with occasional need for information from the database interact using some form of interface, which we call the interactive query interface.

> The precompiler extracts DML commands from an application program written in a host language.

> In the lower part in figure the runtime database processor executes the privileged commands, the executable query plans, and the canned transactions with runtime parameters.

> It works with the system catalog and may update it with statistics.

> The figure shows concurrency control, backup and recovery systems as seperate modules.

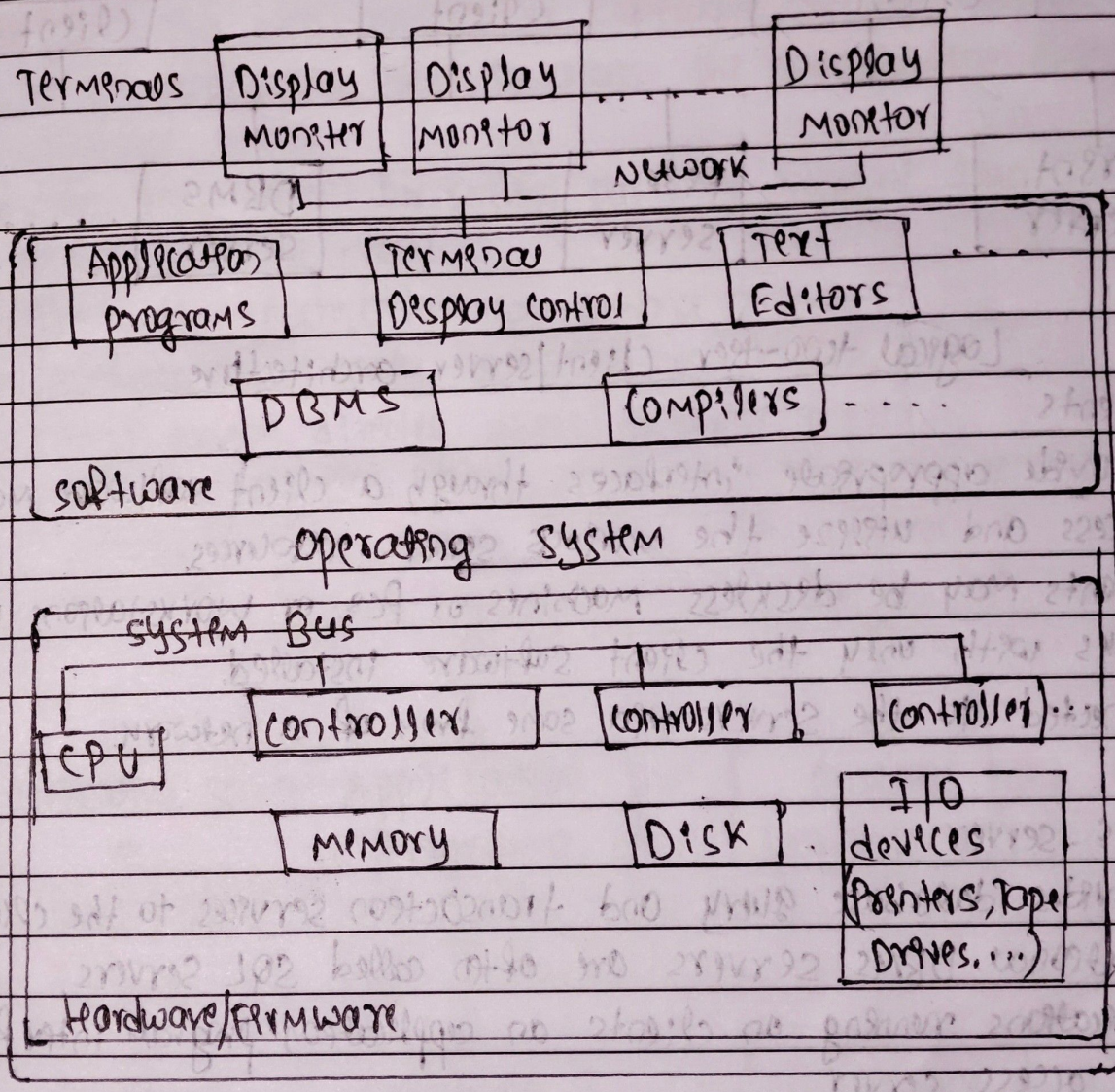
> They are integrated into the working of the runtime database processor for purposes of transaction management.

04) With the need block diagram, explain the architecture of typical DBMS.

→ DBMS architecture

a) Centralized architecture

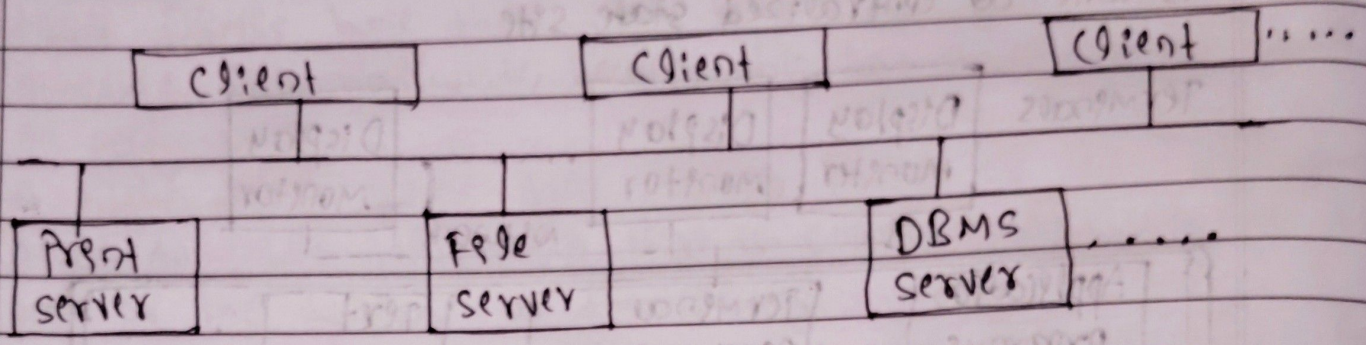
- Combines everything into single system including DBMS software, hardware, application programs and user interface processing software
- User can still connect through a remote terminal - however, all processing is done at centralized ~~state~~ site.



A physical centralized architecture

b) Basic 2-tier client-server Architectures

- specialized server with specialized functions
  - print server
  - File server
  - DBMS server
  - Web server
  - Email server
- clients can access the specialized servers as needed



Logical two-tier client/server architecture

> clients

- Provide appropriate interfaces through a client software module to access and utilize the various servers resources.
- clients may be diskless machines or PCs or workstations with disks with only the client software installed.
- Connected to the servers via some form of a network.

DBMS server

- Provides database query and transaction services to the clients.
- Relational DBMS servers are often called SQL servers.
- Applications running on clients an application program interface (API) to access server.

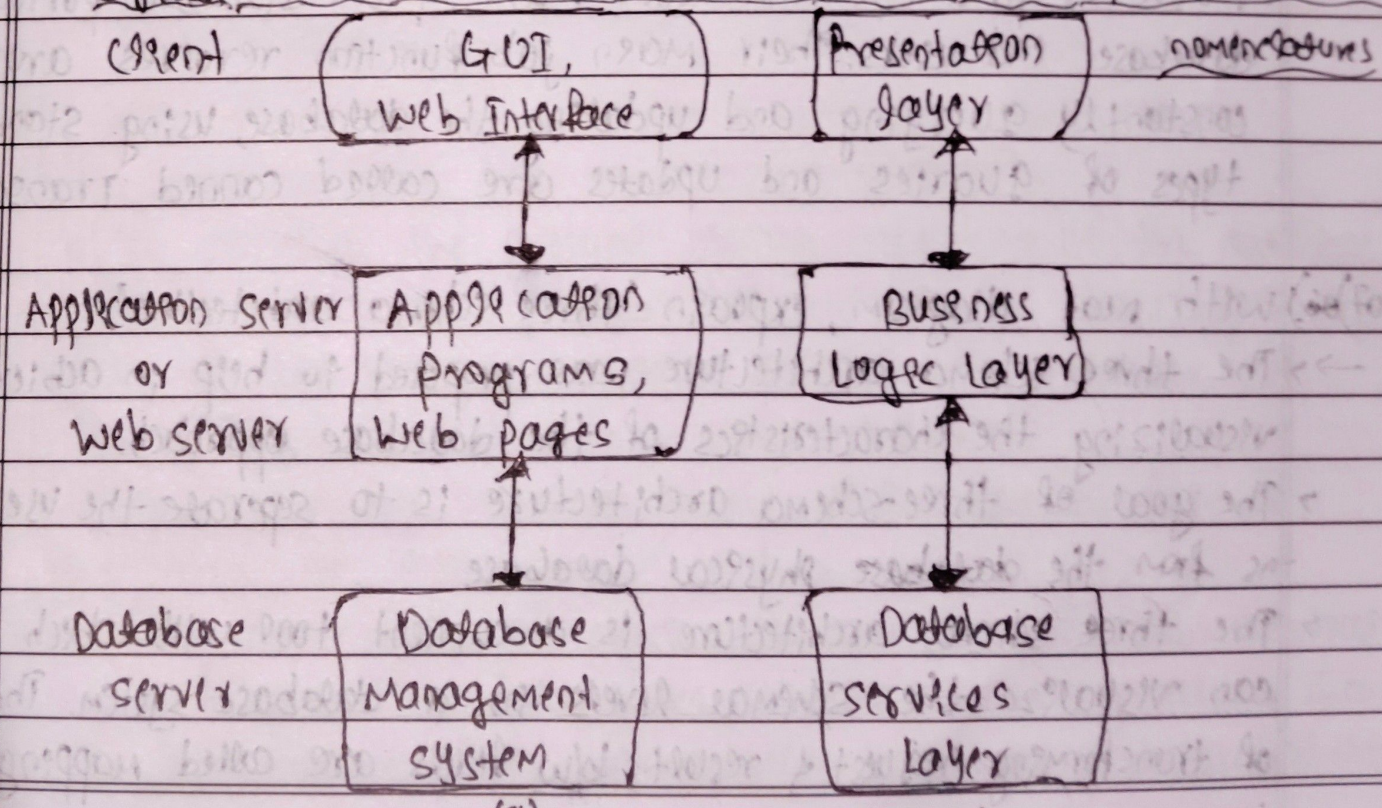
Database via standard interface such as:

- ODBC: open database connectivity standard

- JDBC : For Java Programming access
- A client program may connect to several DBMSs, sometimes called the data sources.
- In general, data sources can be files or other non-DBMS software that manages data.

i) Three-tier client/server Architecture
- Common for web applications
ii) Intermediate Layer called Application server or web server
- Stores the web connectivity software and the business logic part of the application used to access the corresponding data from the database server
- Acts like conduit for sending partially processed data b/w the database server and the client.
iii) Three-tier Architecture can Enhance security:
- Database server only accessible via Middle tier
- Clients cannot directly access database server

Logical three-tier client server architecture with couple of commonly used nomenclatures



(a) (b)

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05) Define the following terms.
i) Data Model ii) Schema iii) Instance iv) Canned Transaction

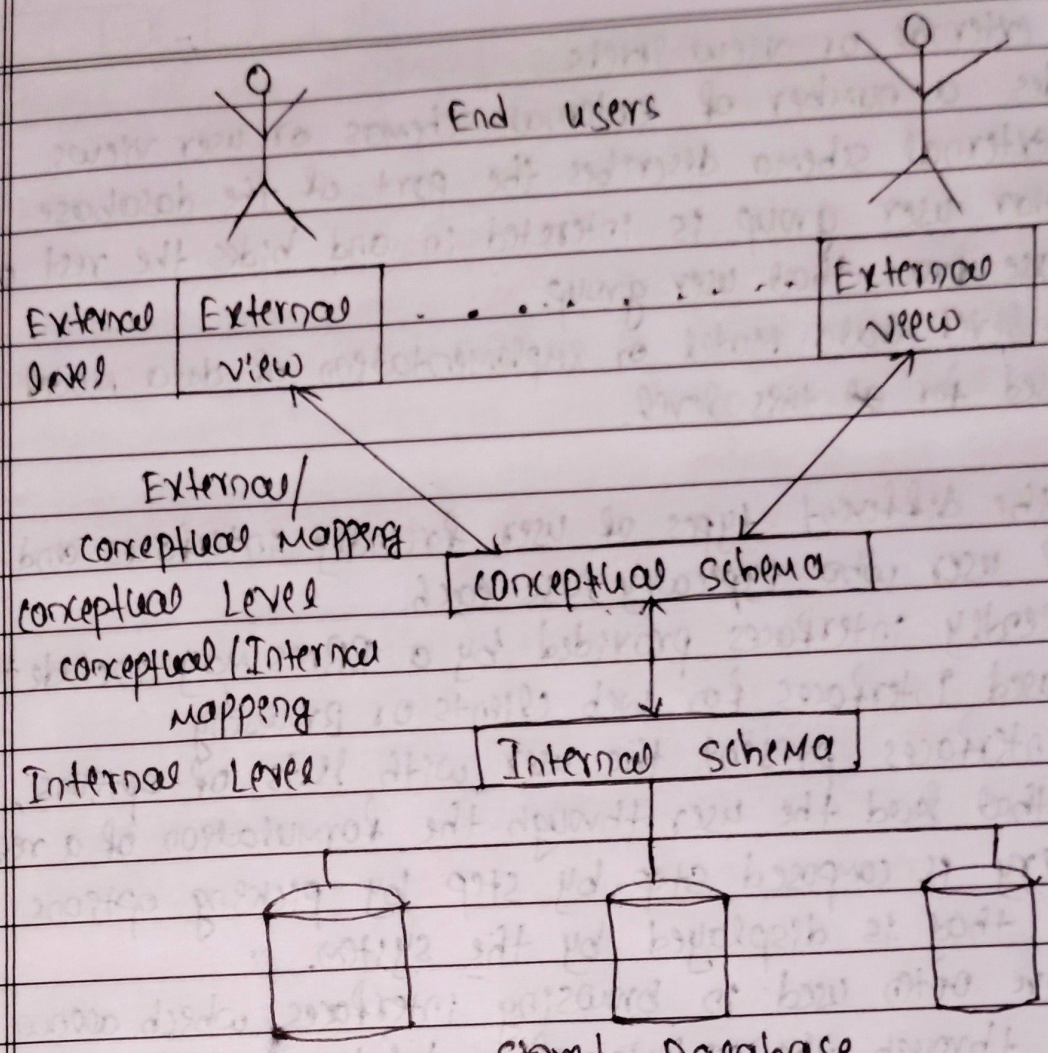
→ i) Data Model
> A data Model is a type of data abstraction that is used to provide the conceptual representation.

ii) Schema
> In any data Model it is important to distinguish between the description of the database and the database itself. The description of a database is called the database schema.

iii) Instance
> The data in the database at a particular moment in time is called a database state or snapshot. It is also called the current set of occurrences or instances in the database.

iv) Canned Transaction
> Naive or parameter end users make up a sizeable portion of database end users. Their main job function revolves around constantly querying and updating the database, using standard types of queries and updates are called canned Transactions.

07) With neat diagram, explain "three schema architecture".
→ > The three schema architecture was proposed to help in achieving and visualizing the characteristics of the database approach.
> The goal of three-schema architecture is to separate the user applications from the physical database.
> The three schema architecture is a convent tool with which the user can visualize the schemas levels in a database system. The process of transforming request & result b/w levels are called Mappings.



The three-schema architecture

> In this architecture schemas can be defined three levels

i) The Internal level has an internal schema
- which describes the physical storage structure of the database.
- Internal schema uses physical data model and describes the complete details of data storage and access paths for the database.

ii) The conceptual level has an conceptual schema
- which describes the structure of the whole database for a community of users.
- It hides the details of physical storage structure and concentrates on describing entities, data types, relationships, user operations & constraints.
- A high level data model or implementation data model can be used at this level.

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iii) The external or view levels
> Includes a number of external schemas or user views.
> Each external schema describes the part of the database that a perticular user group is intersted in and hides the rest of the database from that user group.
> A high level data model or implementation of data model also ~~done~~ used for ~~at~~ this level.

08) Discuss the different types of user friendly interfaces and the types of user who typically use each.
→ User freendly interfaces provided by a DBMS may include the following
i) Menu-Based Interfaces for web clients or Browsing
- These interfaces present the user with lists of options, called menus, that lead the user through the formulation of a request.
- The query is composed step by step by picking options from a menu that is displayed by the system.
- They are often used in browsing interfaces, which allow a user to look through the contents of a database in an exploratory and unstructured manner.

ii) Forms-Based Interfaces
- A forms-based interface displays a form to each user.
- Forms are usually designed and programmed for naive users as interfaces to canned transactions.

iii) Graphical User Interfaces
- A graphical interface (GUI) typically displays a schema to the user in diagrammatic form.
- The user can then specify a query by manipulating the diagram.
- Most GUIs use a pointing device, such as mouse, to pick certain

parts of the displayed schema diagram

iv) Natural Language Interfaces
- These interfaces accept requests written in English or some other language and attempt to "understand" them.
- A natural language interfaces usually has its own "schema," which is similar to the database conceptual schema.
- The natural language interface refers to the words in its schema, as well as to a set of standard words, to interpret the request.

v) Interfaces for Parametric users
- Parametric users, such as bank tellers, often have a small set of operations that they must perform repeatedly.
- Systems analysts and programmers design and implement a special interface for a known class of naive users.

vi) Interfaces for the DBA
- Most database systems contain privileged commands that can be used only by the DBA's staff.
- These includes commands for creating accounts, setting system parameters, granting account authorization, changing a schema, and reorganizing the storage structures of a database.

10) Define the following with an example:
i) DBMS catalog ii) Participation constraints iii) Cardinality ratio
iv) Multi valued attributes

→ i) DBMS catalog
- It is a special database in which definition is stored, which contains information such as the structure of each file, the type & storage format of each data item, & various constraints on the data.

Example of database catalog (relations)

Relation_name	No_of_columns
STUDENT	4
COURSE	4
SECTION	5
GRADE_REPORT	3
PREREQUISITE	2

Columns

Column_name	Data_type	Belongs_to_relation
Name	Character (30)	STUDENT
Student_number	Character (4)	STUDENT
Class	Integer (1)	STUDENT
Major	Major_type	STUDENT
...	...	...
...	...	...
Prerequisite no.		

ii) Participation constraints

> It specifies whether the existence of an entity depends on it's being related to another entity via the relationship type.

> For example, if the company has a rule that each employee must work for exactly one department, then we would like to describe the constraint in the schema. We can distinguish two main types of relationship constraints.

iii) Cardinality Ratio

> Cardinality ratio for a binary relationship specifies the number of relationship instances than an entity can participate in.

> For example, in the WORKS_FOR binary relationship type;

DEPARTMENT : EMPLOYEE is of cardinality ratio 1:N, meaning that each department to numerous employees but an employee can be related to only one department.

iv) Multi valued Attributes

> In some cases an attribute can have a set of values for the same entity is called multi valued attributes

Ex: The colors attribute of a car may have between one and three values, if we assume that a car can have at most three colors.

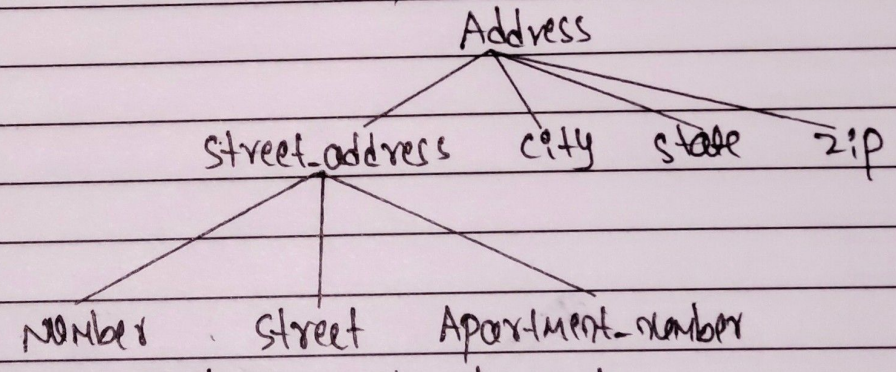
11) Explain different types of attributes that occur in an ER diagram model with example.

→ Serval types of attributes occur in the ER Model

i) Composite attributes

- It can be divided into smaller subparts, which represent more basic attributes with ~~th~~ independent meanings.

> Ex: Address attribute of the employee entity can be sub-divided into streetAddress, City, State and Zip with the values "2311 Kirby," "Houston", "Texas", and "77001".



- This attributes can be hirearchy.
- The value of a composite attribute is the concatenation of the values of its constituent simple attributes.

ii) Simple or atomic attributes

- Attributes that are not divisible are called simple or atomic attributes

EX: Number, street are simple attributes

iii) Single valued attributes

- Most attributes have a single value for a particular entity.

EX: Age is a single-valued attribute of person.

iv) Multivalued attributes

- In some cases an attribute can have a set of values for the same entity

EX: The colors attribute of a car may have b/w one & three values, if we assume that a car at most three colors.

v) Stored versus Derived attributes

- In some cases two attribute values are related - for example, the age and BirthDate attributes of a person.
- The Age attribute is hence called a derived attribute and is said to be derivable from the BirthDate attribute, which is called a stored attribute.

vi) Null values

- In some cases a particular entity may not have an applicable value for an attribute.
- For Example, the ApartmentNumber attribute of an address applies only to addresses that are in apartment buildings and not to other types of residences, such as single-family homes

vii) Complex attributes

- Notice that composite and multivalued attributes can be nested in an arbitary way.

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- We can represent arbitary nesting by grouping components of a composite attribute between parentheses () & seperating the components with commas, and by displaying multivalued attributes between braces {}. Such attributes are called complex attributes.
- For example, PreviousDegrees of a STUDENT is a composite multi-valued attribute denoted by
  {PreviousDegrees (college, year, Degree, Field)}.

12) Define the following with an example.
i) Entity ii) DBMS iii) Attributes iv) Metadata

→ i) Entity

> An Entity represents a real-world object or concept, such as an employee or a project, that is described in the database.

Ex: Student, car, company, university etc.

iii) Attributes

> An attribute represents some property of interest that further describes an entity.

Ex: The employee entity e1 has four attributes: Name, Address, Age and Homephone; their values are "John Smith", "2311 Kirby, Houston, Texas 77001", "55" and "713-749-2630" respectively.

ii) DBMS

> It is a collection of programs that enables users to create and maintain a database.

Ex: Consider an example of an college database system
A college uses a DBMS to store and manage:

- student details (ID, Name, Branch)
- Faculty details
- Course Information

• Marks and attendance

Student_ ID	Student_ Name	Branch es	Year	Course- Id	Course- Name	Faculty Name	Marks	At
101	Rajesh	CSE	2	C101	DBMS	Dr. Kumar	85	90
102	Rahul	ECE	3	C102	DSA	Dr. Sharma	78	85
103	Ankit	ME	1	C103	OS	Dr. Rao	88	95

iv) Metadata

> The information stored in the catalog is called Metadata.

For example if we have a student table:

Student_ID	Name	Branch
101	Rahul	CSE

Its Metadata is

~~Student_ID~~ ~~INT~~ ~~Size~~ ~~Constraint~~

~~Name~~ varchar ~~4~~ ~~primary key~~

~~Branch~~

Field Name	Datatype	Size	Constraint
Student_ID	Int	4	primary key
Name	varchar	50	Not NULL
Branch	varchar	10	-

13) Illustrate Data Independence. Explain the types of data Independence.

→ The structure of data files is stored in the DBMS catalog seperately from the access programs. We call this property program-data independence.

> Two types of data independence

i) Logical and

ii) physical data independence.

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i) Logical data Independence

- It is the capacity to change the conceptual schema without having to change external schemas or application programs
- We may change the conceptual schema to expand the database, or to reduce the database.

ii) Physical data Independence

- It is the capacity to change the internal schema without having to change the conceptual schemas.
- Changes to the internal schema may be needed because some physical files had to be reorganized.
- If the same data as before remains in the database, we should not have to change the conceptual schema.

14) Compare the specialization and generalization with an example.

→ Specialization

- It is the process of defining set of subclasses of an entity type, this entity type is called the superclass of the specialization.
- The set of subclass that forms a specialization is defined on the basis of some distinguishing characteristic of the entities in the superclass.

For example, the set of subclass of SECRETARY, ENGINEER, TECHNICIAN is a specialization of the superclass EMPLOYEE that distinguishes among employee entities based on the job type of each employee.

Generalization

- It is a reverse process of abstraction in which suppresses the differences among serval entity types, identify their common features and generalize them into a single superclass of which the original entity types are special subclasses.

For example, consider the entity types CAR and TRUCK. Because they

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have serval COMMON attributes, they can be generalized into the entity type VEHICLE.

15) Define the following terms:
i) key ii) super key iii) Candidate key iv) Primary key v) Foreign key

→ i) key
> An important constraint on the entities of an entity tye is called as a key.

ii) Super key
> A super key is a set of one or more attributes that are taken collectively and can identify all other attributes uniquely.

iii) Primary key
> A key which is used to uniquely identify each record is known as primary key.

iii) Candidate key
> These are super key which are not having any redundant attributes

v) Foreign key
> It is an attribute or set of attributes in one relation that refers to the primary key of another relation.

16) Define the following:
i) Relation state ii) Relation schema iii) Arity iv) Domain

→ i) Relation state
> A Relation state r of the relation schema $R(A_1, A_2 \ldots A_n)$, also denoted by $r(R)$, is a set of n-tuples $r = \{t_1, t_2 \ldots t_m\}$.

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ii) Relation schema

> A relation schema R, denoted by $R(A_1, A_2, A_3, \ldots A_n)$ is made up of a relation name R and a list of attributes $A_1, A_2 \ldots A_n$.

iii) Arity

> A degree or Arity of a relation is the number of attributes n of its relation schema.

iv) Domain

> A domain D is a set of atomic values.

9) Design an ER-diagram for company database and Relation database schema. Consider minimum 4 entities and indicate cardinality ratio.

-> For company E-R diagram schema

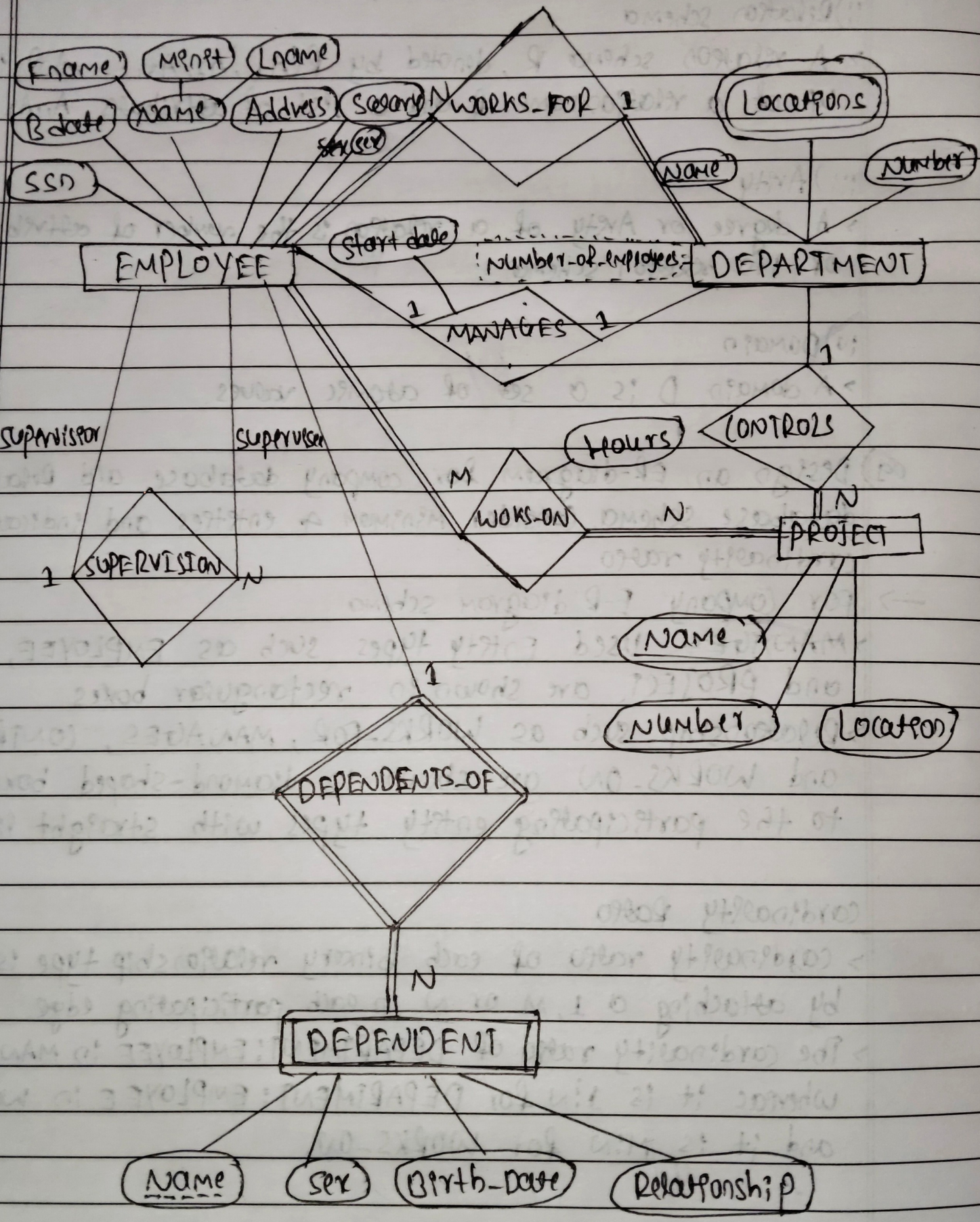
> ~~MANAGES~~ Used Entity types such as EMPLOYEE, DEPARTMENT, and PROJECT are shown in rectangular boxes.

> Relationship such as WORKS_FOR, MANAGES, CONTROLS and WORKS_ON are shown in diamond-shaped boxes attached to the participating entity types with straight lines.

Cardinality Ratio

> cardinality ratio of each binary relationship type is specified by attaching a 1, M or N on each participating edge.

> The cardinality ratio of DEPARTMENT: EMPLOYEE in MANAGES is 1:1, whereas it is 1:N for DEPARTMENT: EMPLOYEE in WORKS_FOR ~~is~~ and it is M:N for WORKS_ON.



E-R schema diagram for COMPANY database